

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	An explanation that includes the following points: • because the same {light source / filter / size cuvette} used for all measurements (1) • because zeroing / calibrating (1) • because objective measurement (1) • because no light lost / gained (1)	described	3

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	An answer that includes the following points : • temperature, use of thermostatically controlled water bath / incubator (1) • pH, use of buffer (1)	accept room	2

Question Number	Answer	Additional Guidance			Mark
2(b)(i)	A table showing the following features: - suitable table drawn (1) - headings with units (1) - all data correctly entered (1)		Loss of betalain from beetroot cells / a.u.		3
		Concentration of salt / mmol dm ⁻³	with ammonium sulfate	with calcium chloride	
		0.00000	0.30	0.30	
		0.00025	0.46	0.29	
		0.00200	0.50	0.15	
		0.02000	0.47	0.02	
		Allow 0.00025 to 0.0003 second cell, first column			

Question Number	Answer	Additional Guidance	Mark
2(b)(iii)	An answer including the following steps : - figures read off graph for CaCl_2 (A) and $(\text{NH}_4)_2\text{SO}_4$ (B), $B-A = C$ (1) $C \div A \times 100 = D$ (%) (1)	allow ecf e.g. $\{0.48 / 0.49\} - 0.09 = \{0.39 / 0.40\}$ accept anything between 0.48 and 0.49 $0.39 \div 0.09 = 433 / 433.3 / 433.33 \%$ $0.40 \div 0.09 = 444 / 444.4 / 444.44 \%$ OR $0.39 \div 0.49 = 80 / 79.6 / 79.59\%$ $0.40 \div 0.49 = 82 / 81.6 / 81.63\%$	correct answer with no working gains 2 marks 2

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	An answer including the following points : • with calcium chloride and ammonium sulfate the {leakage / permeability / intensity of colour} is more than with calcium chloride alone (1) • with ammonium sulfate, leakage increased (compared with water) (1) • {pattern is the same in both lines in second experiment / described} (1) • at 0.010 mmol dm⁻³ calcium chloride alone {leakage / intensity of colour} is {equal / only slightly different} to that with ammonium sulfate with calcium chloride (1) • calcium chloride can reverse the effects of ammonium sulfate (on membranes) (1)	accept increasing calcium chloride conc. leads to decrease in intensity of colour (etc) with or without ammonium sulfate accept correct figures quoted	3

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	An answer including the following points : • repeat readings at 0.002 mmol dm⁻³ (1) • under same conditions (each time) (1) • calculate (mean and) standard deviation (1)	accept error bar accept ref to SD accept perform t-test	3